Studying the learning benefits of GenAI: questions and readings

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Handout for the talk *The challenge of studying a technology's learning benefits* (AI Lab for Research on Teaching and Learning, Tallinn University). Slides and materials: joshuaweidlich.github.io/updates/tallinn-talk.html

Four questions for any study

1. **What exactly was the intervention?** Tool and model version, prompts and instructions, time, teacher support, and what students actually did.
2. **Compared to what?** What the comparison group did in the same time, with the same content and support.
3. **What was learned, and how was it measured?** Knowledge or skill; tested with or without AI access; independent test, researcher-made test, or self-report.
4. **Which claim does the study support?** A programme effect, a design principle, a mechanism, or a description. A claim about the medium as such is usually out of reach.

Further checks

- **Replaceability.** Could another medium, or a person, have delivered the same method? If so, the result is about the method (Clark, 1994).
- **Comparable activity.** Were students doing cognitively comparable things in both groups? Many large effects come from groups that were not (Küchemann et al., 2026).
- **Performance or learning.** Was the test taken with or without AI? Unrestricted GPT-4 raised practice scores by 48% and lowered exam scores without AI by 17% (Bastani et al., 2025). Self-report does not replace a test (Deslauriers et al., 2019).
- **Access or use.** Who used the tool, how, and how much? Did the comparison group use AI too? Log data answers most of this.
- **Expectations.** Did both groups expect the same benefit? Matching time alone leaves expectation effects in place (Boot et al., 2013).
- **Effect size.** In causal education studies with standardised tests, 0.20 SD is already large. Small samples, researcher-made tests and evaluations run by the developer tend to show larger effects (Kraft, 2020; Wolf et al., 2020).
- **Shelf life.** Which model, tested when? Of 50 widely cited claims about LLMs, 44% no longer held on newer models (Queen & Zou, 2026).

- **Informative null.** Would a null result have told us anything? If not, a positive result tells us little about the technology either.

Reading list

To start

- Clark (1983). Reconsidering research on learning from media. *Review of Educational Research*. doi
- Weidlich, Gašević, Drachsler & Kirschner (2025). ChatGPT in education: An effect in search of a cause. *Journal of Computer Assisted Learning*. doi
- Lawson, Martella, Weidlich, Mulders & Buchner (2026). Color me confounded: A critical analysis of media comparisons on ChatGPT in education. *Computers & Education*.
- Bastani et al. (2025). Generative AI without guardrails can harm learning. *PNAS*. doi

Designing and reading studies

- Clark (1994). Media will never influence learning. *ETR&D*. doi
- Mayer (2019). Computer games in education. *Annual Review of Psychology*. doi
- Soderstrom & Bjork (2015). Learning versus performance. *Perspectives on Psychological Science*.
- Boot, Simons, Stothart & Stutts (2013). The pervasive problem with placebos in psychology. *Perspectives on Psychological Science*. doi
- Deslauriers et al. (2019). Measuring actual learning versus feeling of learning. *PNAS*. doi
- Kraft (2020). Interpreting effect sizes of education interventions. *Educational Researcher*.
- Wolf et al. (2020). Average effect sizes in developer-commissioned and independent evaluations. *Journal of Research on Educational Effectiveness*. doi

State of the evidence

- Bartoš, Bujak, Martinková & Wagenmakers (2026). Effect of artificial intelligence on learning: A meta-meta-analysis. Preprint. doi
- Küchemann et al. (2026). Evidence of impact and interpretational limits of generative AI in STEM education. *Artificial Intelligence Review*. doi
- O'Neill (2026). Presumed effective: The manufacturing of an evidence base for AI-in-education through flawed meta-analysis. Preprint, EdArXiv.
- Queen & Zou (2026). The shelf life of LLM findings. Website